

use assignment;

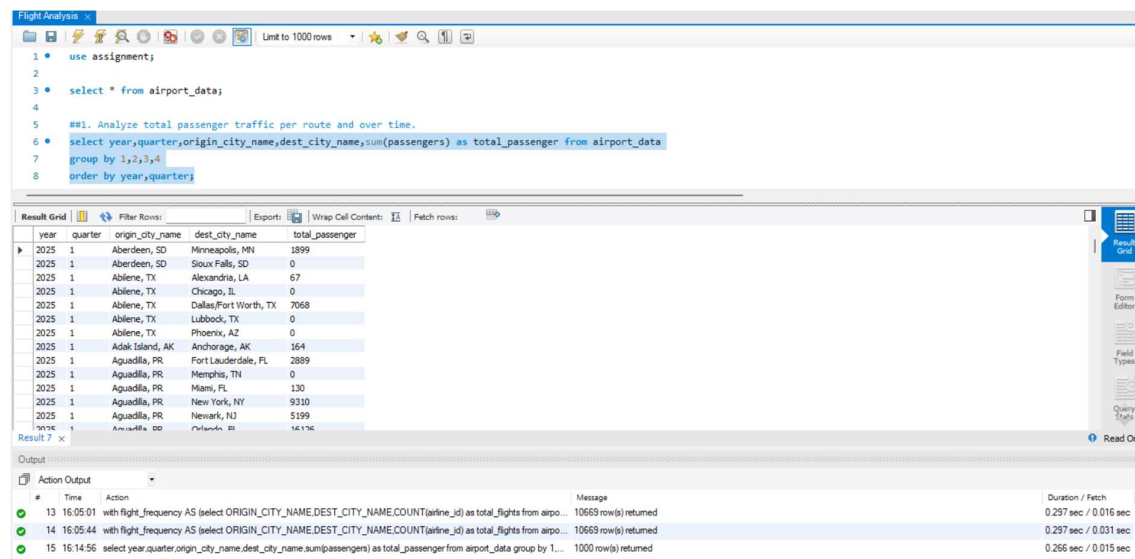
select * from airport_data;

##1. Analyze total passenger traffic per route and over time.

select year,quarter,origin_city_name,dest_city_name,sum(passengers) as
total_passenger from airport_data

group by 1,2,3,4

order by year,quarter;



The screenshot shows a SQL IDE window titled "Flight Analysis". The query editor contains the following SQL code:

```
1 use assignment;
2
3 select * from airport_data;
4
5 ##1. Analyze total passenger traffic per route and over time.
6 select year,quarter,origin_city_name,dest_city_name,sum(passengers) as total_passenger from airport_data
7 group by 1,2,3,4
8 order by year,quarter;
```

The "Result Grid" shows the following data:

year	quarter	origin_city_name	dest_city_name	total_passenger
2025	1	Aberdeen, SD	Minneapolis, MN	1899
2025	1	Aberdeen, SD	Sioux Falls, SD	0
2025	1	Ablene, TX	Alexandria, LA	67
2025	1	Ablene, TX	Chicago, IL	0
2025	1	Ablene, TX	Dallas/Fort Worth, TX	7068
2025	1	Ablene, TX	Lubbock, TX	0
2025	1	Ablene, TX	Phoenix, AZ	0
2025	1	Adak Island, AK	Anchorage, AK	164
2025	1	Aguadilla, PR	Fort Lauderdale, FL	2889
2025	1	Aguadilla, PR	Memphis, TN	0
2025	1	Aguadilla, PR	Miami, FL	130
2025	1	Aguadilla, PR	New York, NY	9310
2025	1	Aguadilla, PR	Newark, NJ	5199
2025	1	Aguadilla, PR	Orlando, FL	14,174

The "Output" pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
13	16:05:01	with flight_frequency AS (select ORIGIN_CITY_NAME,DEST_CITY_NAME,COUNT(airline_id) as total_flights from airport_data)	10669 row(s) returned	0.297 sec / 0.016 sec
14	16:05:44	with flight_frequency AS (select ORIGIN_CITY_NAME,DEST_CITY_NAME,COUNT(airline_id) as total_flights from airport_data)	10669 row(s) returned	0.297 sec / 0.031 sec
15	16:14:56	select year,quarter,origin_city_name,dest_city_name,sum(passengers) as total_passenger from airport_data group by 1,2,3,4 order by year,quarter;	1000 row(s) returned	0.266 sec / 0.015 sec

##2. Determine average passengers per flight for various routes and airports.

##For Routes

select ORIGIN_CITY_NAME,DEST_CITY_NAME,airline_id,avg(passengers) as
average_passenger from airport_data

group by 1,2,3

order by average_passenger desc;

Flight Analysis

Limit to 1000 rows

```

8 order by year,quarter;
9
10 ##2. Determine average passengers per flight for various routes and airports.
11 ##For Routes
12 select ORIGIN_CITY_NAME,DEST_CITY_NAME,airline_id,avg(passengers) as average_passenger from airport_data
13 group by 1,2,3
14 order by average_passenger desc;
15

```

Result Grid

ORIGIN_CITY_NAME	DEST_CITY_NAME	airline_id	average_passenger
Orlando, FL	Atlanta, GA	19790	81227
Tampa, FL	Atlanta, GA	19790	61271
Kahului, HI	Honolulu, HI	19690	59550
Fort Lauderdale, FL	Atlanta, GA	19790	59507
Miami, FL	Dallas/Fort Worth, TX	19805	58967
Honolulu, HI	Kahului, HI	19690	58501
Houston, TX	Denver, CO	19977	58424
Atlanta, GA	Tampa, FL	19790	57115
Orlando, FL	Charlotte, NC	19805	55684
Dallas/Fort Worth, TX	Miami, FL	19805	54948
Dallas/Fort Worth, TX	Los Angeles, CA	19805	54048
Dallas/Fort Worth, TX	Phoenix, AZ	19805	53516
Atlanta, GA	Fort Lauderdale, FL	19790	53099
Dallas/Fort Worth, TX	Dallas/Fort Worth, TX	19805	47008

Output

#	Time	Action	Message	Duration / Fetch
14	16:05:44	with flight_frequency AS (select ORIGIN_CITY_NAME,DEST_CITY_NAME,COUNT(airline_id) as total_flights from airport_data)	10669 row(s) returned	0.297 sec / 0.031 sec
15	16:14:56	select year,quarter,origin_city_name,dest_city_name,sum(passengers) as total_passenger from airport_data group by 1,2,3	1000 row(s) returned	0.266 sec / 0.015 sec
16	16:16:07	select ORIGIN_CITY_NAME,DEST_CITY_NAME,airline_id,avg(passengers) as average_passenger from airport_data group by 1,2,3	1000 row(s) returned	0.125 sec / 0.000 sec

##For airports

with outgoing_passenger as

(select ORIGIN_AIRPORT_ID, airline_id, avg(passengers) as average_passenger from airport_data

group by 1,2

order by average_passenger desc),

incoming_passenger as

(select DEST_AIRPORT_ID, airline_id, avg(passengers) as average_passenger from airport_data

group by 1,2

order by average_passenger desc),

all_airport as

(select DISTINCT ORIGIN_AIRPORT_ID AS airport_id from airport_data

UNION

select DISTINCT DEST_AIRPORT_ID AS airport_id from airport_data)

```
select a.airport_id,op.average_passenger + ip.average_passenger as total_passenger  
from all_airport as a
```

```
LEFT JOIN outgoing_passenger as op on a.airport_id = op.ORIGIN_AIRPORT_ID
```

```
LEFT JOIN incoming_passenger as ip on a.airport_id = ip.DEST_AIRPORT_ID
```

```
where op.average_passenger + ip.average_passenger is not null
```

```
order by total_passenger desc;
```

The screenshot shows a SQL query editor with the following query:

```
16 --For airports  
17 with outgoing_passenger as  
18 (select ORIGIN_AIRPORT_ID, airline_id, avg(passengers) as average_passenger from airport_data  
19 group by 1,2  
20 order by average_passenger desc),  
21  
22 incoming_passenger as  
23 (select DEST_AIRPORT_ID, airline_id, avg(passengers) as average_passenger from airport_data  
24 group by 1,2  
25 order by average_passenger desc),  
26
```

The result grid shows the following data:

airport_id	total_passenger
13204	50551.34444444444
12402	41786
13931	40376
10599	38388
14193	38029
13204	37328.49707602339
12889	37303.55555555556
13204	37149.740740740745
15304	36609.5
13204	36414.066666666666
12402	35987
13204	35610.34920634921
13931	35276.666666666664
12402	35056
13204	34294.4269005848
13256	33911
10397	33704.444444444444

-- 3. Assess flight frequency and identify high-traffic corridors.

```
select ORIGIN_CITY_NAME,DEST_CITY_NAME,count(airline_id) as total_flights from  
airport_data
```

```
group by 1,2
```

```
order by total_flights desc;
```

Flight Analysis

```

37
38 -- 3. Assess flight frequency and identify high-traffic corridors.
39 * select ORIGIN_CITY_NAME,DEST_CITY_NAME,count(airline_id) as total_flights from airport_data
40 group by 1,2
41 order by total_flights desc;

```

ORIGIN_CITY_NAME	DEST_CITY_NAME	total_flights
Seattle, WA	Los Angeles, CA	16
Dallas/Fort Worth, TX	Houston, TX	15
New York, NY	Chicago, IL	14
Los Angeles, CA	Seattle, WA	14
Chicago, IL	St. Louis, MO	13
Chicago, IL	Cincinnati, OH	13
Las Vegas, NV	Los Angeles, CA	13
Columbus, OH	Atlanta, GA	13
Chicago, IL	New York, NY	13
Houston, TX	Dallas/Fort Worth, TX	13
Los Angeles, CA	Las Vegas, NV	12
Chicago, IL	Minneapolis, MN	12
Atlanta, GA	Columbus, OH	12
Washington, DC	Detroit, MI	12
San Francisco, CA	Los Angeles, CA	12
Los Angeles, CA	Phoenix, AZ	12
Detroit, MI	Chicago, IL	12
Chicago, IL	Detroit, MI	12
Minneapolis, MN	Washington, DC	11
Detroit, MI	Washington, DC	11
Washington, DC	Providence, RI	11
Atlanta, GA	Washington, DC	11
Orlando, FL	Miami, FL	11
Washington, DC	Salisbury/Bohannon, NC	11

Result 10

-- 4. Compare passenger numbers across origin cities to identify top-performing airports.

```
select ORIGIN_CITY_NAME,ORIGIN_AIRPORT_ID,sum(passengers) as total_passenger
from airport_data
```

```
group by 1,2
```

```
order by total_passenger desc;
```

Flight Analysis

```

43
44 -- 4. Compare passenger numbers across origin cities to identify top-performing airports.
45 * select ORIGIN_CITY_NAME,ORIGIN_AIRPORT_ID,sum(passengers) as total_passenger from airport_data
46 group by 1,2
47 order by total_passenger desc;

```

ORIGIN_CITY_NAME	ORIGIN_AIRPORT_ID	total_passenger
Atlanta, GA	10397	3095254
Denver, CO	11292	2728395
Dallas/Fort Worth, TX	11298	2575266
Chicago, IL	13930	2139178
Orlando, FL	13204	2001905
Las Vegas, NV	12889	1969324
Phoenix, AZ	14107	1811982
Charlotte, NC	11057	1793744
Los Angeles, CA	12892	1774710
Seattle, WA	14747	1502472
Miami, FL	13303	1341597
Houston, TX	12266	1302628
San Francisco, CA	14771	1296640
Newark, NJ	11618	1293553
Fort Lauderdale, FL	11697	1180673
New York, NY	12953	1118522
Minneapolis, MN	13487	1108111
Boston, MA	10721	1070537
New York, NY	12478	1014489
Salt Lake City, UT	14869	994768
Detroit, MI	11433	981699
Tampa, FL	15304	880956
San Diego, CA	14679	862613
Philadelphia, PA	14100	847884

Result 11

-- 5. Evaluate available seat capacity to understand seat utilization.

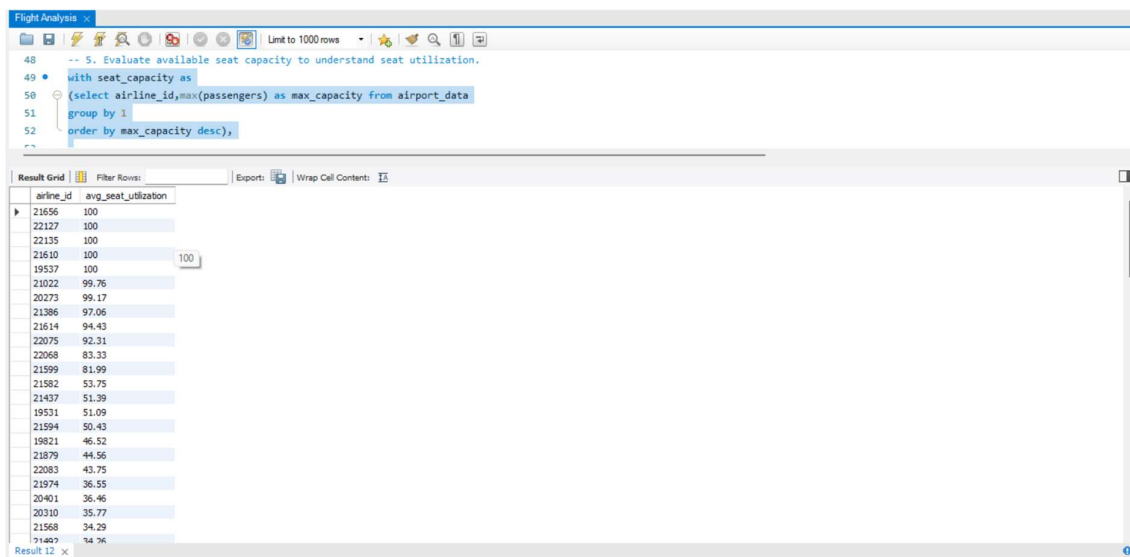
with seat_capacity as

```
(select airline_id,max(passengers) as max_capacity from airport_data  
group by 1  
order by max_capacity desc),
```

seat_utilization as

```
(select a.airline_id, a.passengers*100/sc.max_capacity as used_seats from  
airport_data a  
join seat_capacity sc on a.airline_id = sc.airline_id  
order by used_seats desc)
```

```
select airline_id,round(avg(used_seats),2) as avg_seat_utilization from seat_utilization  
group by 1  
order by avg_seat_utilization desc;
```



The screenshot shows a SQL IDE window titled "Flight Analysis". The query editor contains the following SQL code:

```
-- 5. Evaluate available seat capacity to understand seat utilization.  
48  
49 with seat_capacity as  
50 (select airline_id,max(passengers) as max_capacity from airport_data  
51 group by 1  
52 order by max_capacity desc),
```

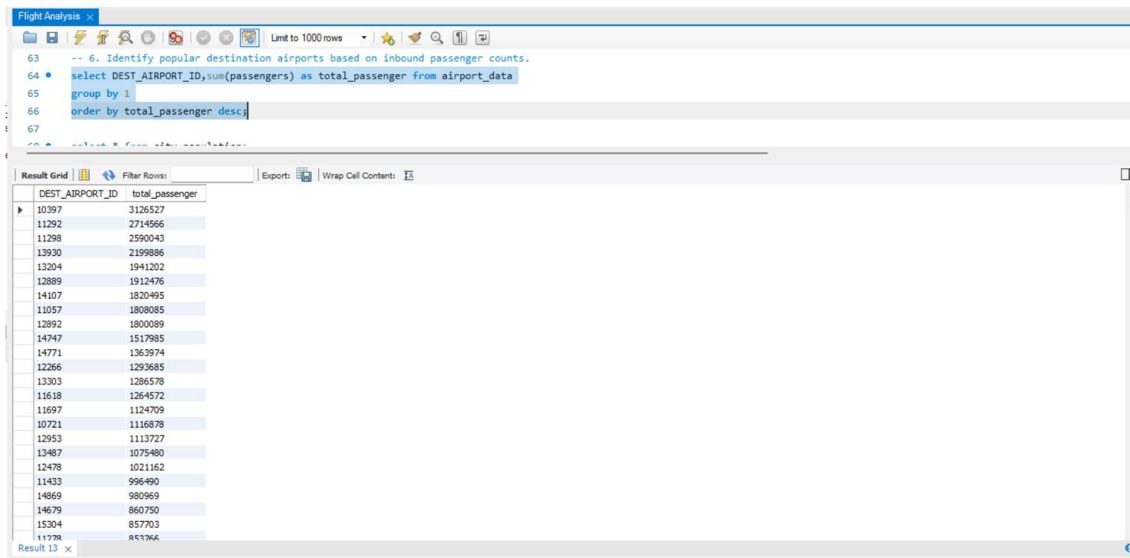
The results pane shows a table with two columns: "airline_id" and "avg_seat_utilization". The table contains 25 rows of data, sorted by "avg_seat_utilization" in descending order. The first row has an "airline_id" of 21656 and an "avg_seat_utilization" of 100. The last row has an "airline_id" of 19409 and an "avg_seat_utilization" of 14.74.

airline_id	avg_seat_utilization
21656	100
22127	100
22135	100
21610	100
19537	100
21022	99.76
20273	99.17
21386	97.06
21614	94.43
22075	92.31
22068	83.33
21999	81.99
21582	53.75
21437	51.39
19531	51.09
21594	50.43
19821	46.52
21879	44.56
22083	43.75
21974	36.55
20401	36.46
20310	35.77
21568	34.29
19409	14.74

-- 6. Identify popular destination airports based on inbound passenger counts.

```
select DEST_AIRPORT_ID,sum(passengers) as total_passenger from airport_data  
group by 1  
order by total_passenger desc;
```

```
select * from city_population;
```



The screenshot shows a SQL IDE window titled "Flight Analysis". The query editor contains the following SQL code:

```
-- 6. Identify popular destination airports based on inbound passenger counts.  
select DEST_AIRPORT_ID, sum(passengers) as total_passenger from airport_data  
group by 1  
order by total_passenger desc;
```

The results grid displays the following data:

DEST_AIRPORT_ID	total_passenger
10397	3126527
11292	2714566
11298	2590043
13930	2199886
13204	1941202
12889	1912476
14107	1820495
11057	1806085
12892	1800089
14747	1517985
14771	1363974
12266	1293685
13303	1286578
11618	1264572
11697	1124709
10721	1116878
12953	1113727
13487	1075480
12478	1021162
11433	996490
14869	980969
14679	860750
15304	857703
11578	847764

```
-- 7. Examine the relationship between city population and airport passenger traffic.
```

with outgoing_passenger as

```
(select ORIGIN_AIRPORT_ID, avg(passengers) as avg_passenger from airport_data
```

```
group by 1
```

```
order by avg_passenger desc),
```

incoming_passenger as

```
(select DEST_AIRPORT_ID, avg(passengers) as avg_passenger from airport_data
```

```
group by 1
```

```
order by avg_passenger desc),
```

airports as

```
(select distinct ORIGIN_AIRPORT_ID as airport_id, ORIGIN_CITY_NAME AS city_name  
from airport_data
```

```
UNION
```

```
select distinct DEST_AIRPORT_ID as airport_id, DEST_CITY_NAME AS city_name from  
airport_data),
```

passenger_traffic as

(select a.city_name, ip.avg_passenger + op.avg_passenger as
total_passenger_travelling from airports as a

LEFT JOIN incoming_passenger as ip on a.airport_id=ip.DEST_AIRPORT_ID

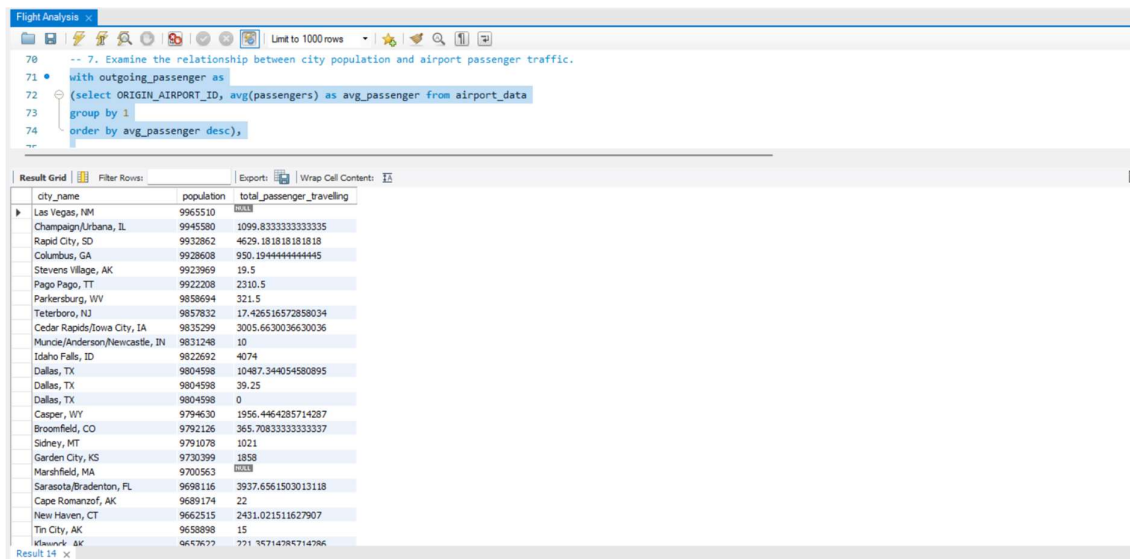
LEFT JOIN outgoing_passenger as op on a.airport_id=op.ORIGIN_AIRPORT_ID

order by total_passenger_travelling desc)

select c.city_name,c.population,p.total_passenger_travelling from passenger_traffic as
p

join city_population as c on p.city_name=c.city_name

order by population desc;



```
-- 7. Examine the relationship between city population and airport passenger traffic.
with outgoing_passenger as
(select ORIGIN_AIRPORT_ID, avg(passengers) as avg_passenger from airport_data
group by 1
order by avg_passenger desc),
```

city_name	population	total_passenger_travelling
Las Vegas, NM	9965510	10000
Champaign/Urbana, IL	9945580	1099.8333333333335
Rapid City, SD	9932862	4629.181818181818
Columbus, GA	9928608	950.1944444444444
Stevens Village, AK	9925969	19.5
Pago Pago, TT	9922208	2310.5
Parkersburg, WV	9858694	321.5
Teterboro, NJ	9857832	17.426516572858034
Cedar Rapids/Iowa City, IA	9835299	3005.6630036630036
Muncie/Anderson/Newcastle, IN	9831248	10
Idaho Falls, ID	9822692	4074
Dallas, TX	9804598	10487.344054580895
Dallas, TX	9804598	39.25
Dallas, TX	9804598	0
Casper, WY	9794630	1956.4464285714287
Broomfield, CO	9792126	365.70833333333337
Sidney, MT	9791078	1021
Garden City, KS	9730399	1858
Marshfield, MA	9700563	10000
Sarasota/Bradenton, FL	9698116	3937.6561503013118
Cape Romanzof, AK	9689174	22
New Haven, CT	9662515	2431.021511627907
Tin City, AK	9658898	15
Wausau, WI	9647677	771.9571428571429

-- 8. Assess the impact of population size on flight frequency and route choices.

select * from airport_data;

select * from city_population;

with flight_frequency AS

(select ORIGIN_CITY_NAME,DEST_CITY_NAME,COUNT(airline_id) as total_flights from
airport_data

group by 1,2

order by total_flights desc)

```
select f.total_flights,f.ORIGIN_CITY_NAME, f.DEST_CITY_NAME, cp1.population +  
cp2.population as total_population from flight_frequency f
```

```
left join city_population cp1 on f.ORIGIN_CITY_NAME=cp1.city_name
```

```
left join city_population cp2 on f.DEST_CITY_NAME=cp2.city_name
```

```
order by f.total_flights,total_population desc;
```

Flight Analysis

Limit to 1000 rows

```
96 -- 8. Assess the impact of population size on flight frequency and route choices.  
97 select * from airport_data;  
98 select * from city_population;  
99  
100 with flight_frequency AS
```

Result Grid

	total_flights	ORIGIN_CITY_NAME	DEST_CITY_NAME	total_population
1	1	Dallas, TX	Las Vegas, NM	19770108
1	1	Teterboro, NJ	Marshfield, MA	19558395
1	1	Sarasota/Bradenton, FL	Cedar Rapids/Iowa City, IA	19533415
1	1	Cedar Rapids/Iowa City, IA	Sarasota/Bradenton, FL	19533415
1	1	New Haven, CT	Teterboro, NJ	19520347
1	1	Dallas, TX	Sarasota/Bradenton, FL	19502714
1	1	Sarasota/Bradenton, FL	Dallas, TX	19502714
1	1	Dallas, TX	Sacramento, CA	19411817
1	1	Sacramento, CA	Dallas, TX	19411817
1	1	Sarasota/Bradenton, FL	New Haven, CT	19360631
1	1	New Haven, CT	Sarasota/Bradenton, FL	19360631
1	1	Chicopee Falls, MA	New Haven, CT	19289891
1	1	New Haven, CT	Chicopee Falls, MA	19289891
1	1	Champaign/Urbana, IL	Lincoln, NE	19271821
1	1	Lincoln, NE	Champaign/Urbana, IL	19271821
1	1	Hancock/Houghton, MI	Escanaba, MI	19192523
1	1	Sarasota/Bradenton, FL	Syracuse, NY	19183948
1	1	Syracuse, NY	Sarasota/Bradenton, FL	19183948
1	1	Sacramento, CA	Kona, HI	19169926
1	1	Kona, HI	Sacramento, CA	19169926
1	1	Cedar Rapids/Iowa City, IA	Lincoln, NE	19161540
1	1	Lincoln, NE	Cedar Rapids/Iowa City, IA	19161540
1	1	Dallas, TX	Philadelphia, PA	19143345
1	1	Philadelphia, PA	Dallas, TX	19143345

Result 15 x

```
with seat_capacity as (select airline_id,max(passengers) as max_capacity from
airport_data group by 1 order by max_capacity desc), seat_utilization as (select
a.airline_id, a.passengers*100/sc.max_capacity as used_seats from airport_data a join
seat_capacity sc on a.airline_id = sc.airline_id order by used_seats desc) select
airline_id,round(avg(seat_utilization),2) as avg_seat_utilization from seat_utilization
group by 1 order by avg_seat_utilization desc
```